

**EGERTON**



**UNIVERSITY**

**UNIVERSITY EXAMINATIONS**

**REGULAR -NJORO CAMPUS**

**SECOND SEMESTER, 2019/2020 ACADEMIC YEAR**

**FIRST YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE**

**IN ENVIRONMENTAL SCIENCE**

**ENSC 306: ENVIRONMENTAL HEALTH**

**STREAM: BSc. (ENSC & NARE)**

**TIME: 2 HRS**

**EXAMINATION SESSION: APRIL**

**YEAR: 2021**

**INSTRUCTIONS:**

- (i) Attempt all questions in Section A and any TWO in Section B
- (ii) Marks for each question are indicated in parenthesis
- (iii) Do not write on the question paper
- (iv) Illustrate your answers with suitable diagrams and formulas where appropriate

**SECTION A (40 MARKS)**

**Question One**

Write short notes on the following terms as used in environmental health.

- a) The inner versus the outer environment **(3 Marks)**
- b) Chemical interactions **(4 Marks)**
- c) Selective toxicity **(3 Marks)**

**Question Two**

- a) Many factors impact sensitivity to a pollutant in human beings. Explain these factors and give examples in each case. **(4 Marks)**
- b) What do you understand by environmental hormones? Discuss the impacts of environmental hormones to fish and other wildlife **(6 Marks)**

**Question Three**

The skin acts as a protective barrier separating the rest of the body from the outside environment. In the light of this statement, discuss the functions of the parts of the skin and how chemicals are absorbed into the bloodstream. (10 Marks)

**Question Four**

Toxicants can affect any tissue or organ within the body. Discuss the effects of toxic substances to human liver. (10 Marks)

**SECTION B (30 MARKS)**

**Question Five**

"The specific response that an organism exhibits from a given dose of a toxin varies". Discuss. (15 Marks)

**Question Six**

There are common toxins introduced into food through improper handling. Discuss the main organisms that contaminate food as a result of improper handling. (15 Marks)

**Question Seven**

Dose-response assessment is an important step in risk assessment with respect to environmental toxicants. Discuss the various steps to be considered in dose-response assessment. (15 Marks)

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